

EX NO:1 Butterworth IIR Filters

1.1 Design and Analysis of a Butterworth IIR Low-Pass Filter Using MATLAB.

Program :

```
clc; clear
```

```
Fs=10000; Fp=1000; Fst=1500;
```

```
[N,Wn]=buttord(Fp/(Fs/2),Fst/(Fs/2),1,60);
```

```
[b,a]=butter(N,Wn);
```

```
freqz(b,a) % Magnitude & Phase
```

```
figure
```

```
[h,n]=impz(b,a,50);
```

```
...title('Impulse Response')
```

```
figure
```

```
stepz(b,a)
```

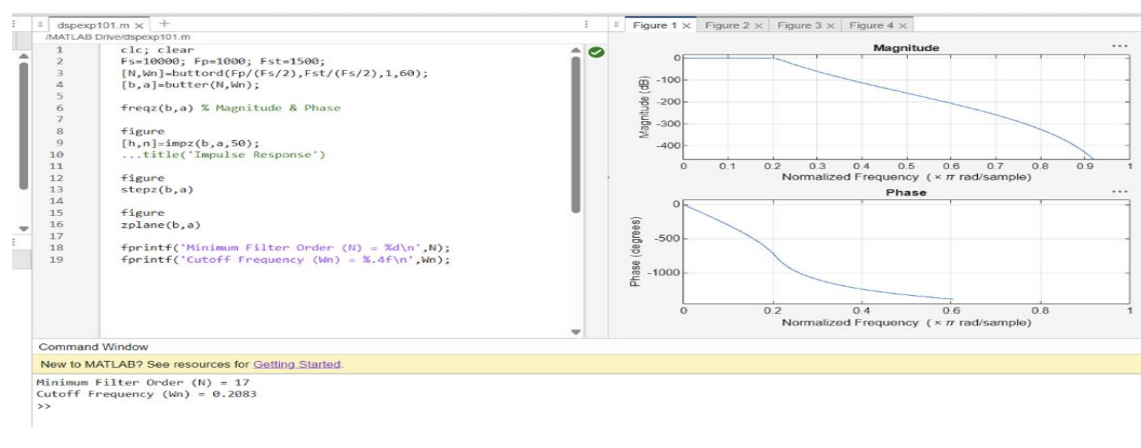
```
figure
```

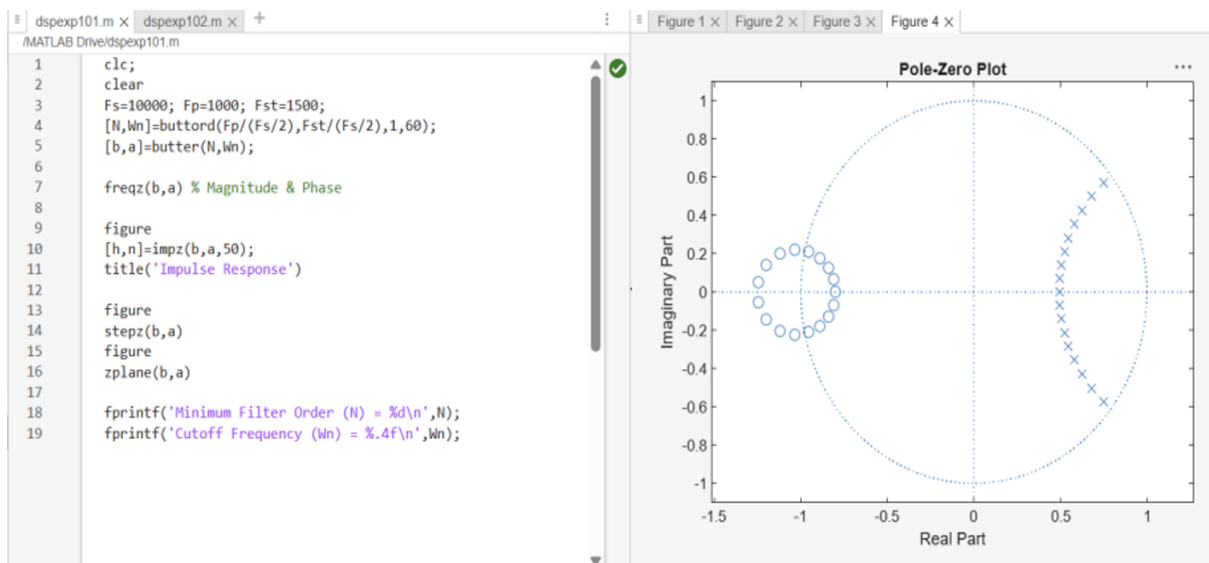
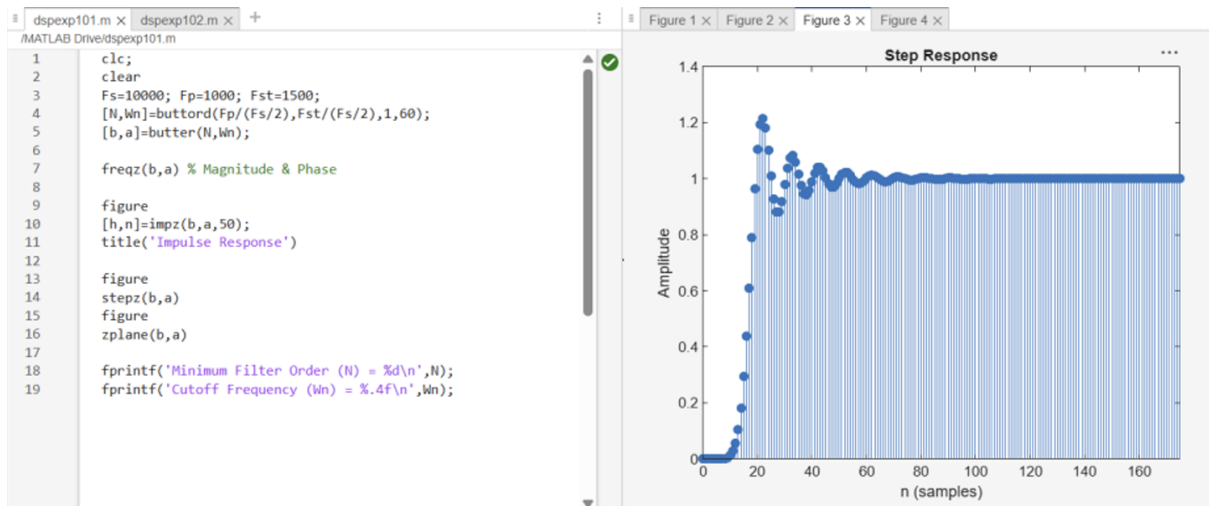
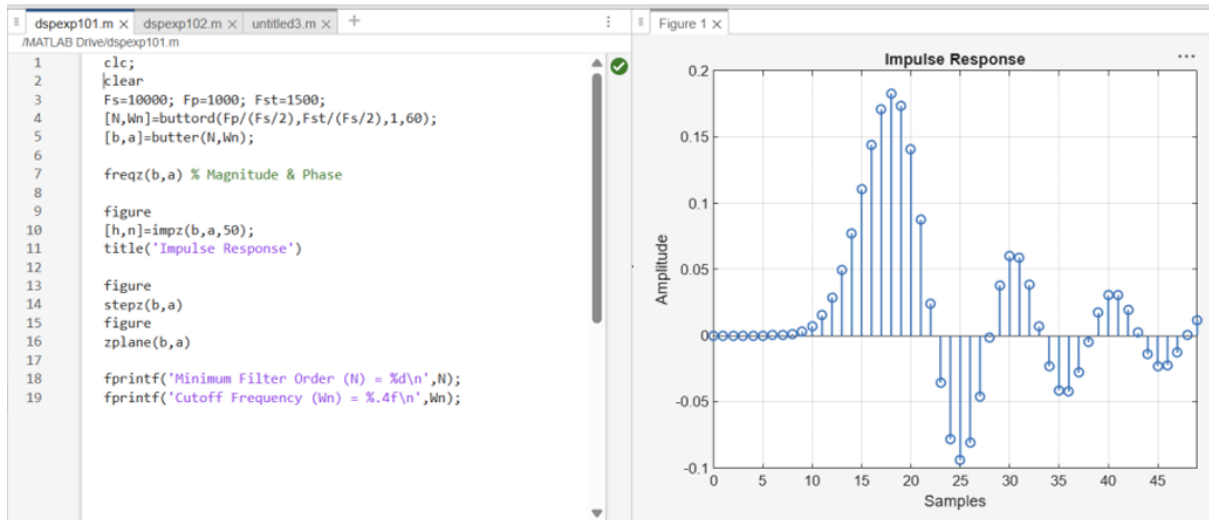
```
zplane(b,a)
```

```
fprintf('Minimum Filter Order (N) = %d\n',N);
```

```
fprintf('Cutoff Frequency (Wn) = %.4f\n',Wn);
```

Output :





1.2. Design and Analysis of a Butterworth IIR High-Pass Filter Using MATLAB.

Program :

```
clc; clear

Fs=10000; Fp=3000; Fst=2000;

[N,Wn]=buttord(Fp/(Fs/2),Fst/(Fs/2),1,60);

[b,a]=butter(N,Wn,'high');

% Magnitude & Phase

freqz(b,a)

% Impulse Response

figure

[h,n]=impz(b,a,50);

stem(n,h), grid on

title('Impulse Response')

% Step Response

figure

stepz(b,a)

title('Step Response')

% Group Delay

figure

grpdelay(b,a)

title('Group Delay')

% Pole-Zero Plot

figure

zplane(b,a)
```

title('Pole-Zero Plot')

fprintf('Minimum Filter Order (N) = %d\n',N);

fprintf('Cutoff Frequency (Wn) = %.4f\n',Wn);

Output :

